

43. For each of the following real matrices, determine the characteristic polynomial, the eigenvalues with their algebraic and geometric multiplicities, and a basis for each eigenspace.

Demonstrate how to compute the characteristic polynomial and eigenvalues in two ways:

- (1) Compute the characteristic polynomial directly in expanded form by applying the Triangle Rule or Sarrus' Rule, then use the theorem on integer roots to factor it.
- (2) Use suitable row/column operations to factor out a term of the form $(t - \lambda)$ at an early stage of the computation (as shown in the lecture): this simplifies the factorization substantially.

$$A = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix}.$$

44. For each of the following real matrices, determine the eigenvalues with their algebraic and geometric multiplicities, and a basis for each eigenspace.

Do not compute the characteristic polynomial directly! Instead, use relevant properties and propositions, such as eigenvalues of non-invertible matrices, the sum and the product of eigenvalues, $\text{GM} \leq \text{AM}$, etc.

$$A = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 6 & 0 \\ 1 & 0 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 5 & 5 & 5 \\ 7 & 7 & 7 & 7 \\ 11 & 11 & 11 & 11 \\ -23 & -23 & -23 & -23 \end{pmatrix}.$$

45. Let $A, B \in F^{n \times n}$. Prove the following claims.

- (a) If λ is an eigenvalue of A and $c \in F$, then $\lambda + c$ is an eigenvalue of $A + cI$.
- (b) If λ is an eigenvalue of A , then λ^2 is an eigenvalue of A^2 .
- (c) If A is invertible and λ is an eigenvalue of A , then λ^{-1} is an eigenvalue of A^{-1} .
- (d) If v is an eigenvector of both A and B , then v is also an eigenvector of AB .

46. NOT FOR SUBMISSION

For the following matrices, determine the characteristic polynomial, the eigenvalues with algebraic and geometric multiplicities, and a basis for every eigenspace.

- For the following four matrices, apply the standard method: compute the characteristic polynomial, etc. Note that in some cases it is possible to factor out one of the terms $(t - \lambda)$ when expanding $\det(A - tI)$, as in Question 43.

$$\begin{pmatrix} 3 & -5 & 5 \\ 2 & -7 & 8 \\ 2 & -5 & 6 \end{pmatrix}, \quad \begin{pmatrix} -1 & 3 & 3 \\ 6 & -4 & -6 \\ -3 & 3 & 5 \end{pmatrix}, \quad \begin{pmatrix} 2 & 1 & 1 \\ -1 & 0 & -1 \\ 1 & 2 & 3 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{pmatrix}.$$

- For the following four matrices, use propositions about eigenvalues and eigenvectors, as in Question 44.

$$\begin{pmatrix} -2 & 7 & 1 \\ 0 & 2 & 0 \\ 4 & -7 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 0 \\ -2 & -2 & -2 \\ 0 & 0 & 1 \end{pmatrix}.$$

- For the following two matrices, computing the eigenvectors corresponding to nonzero eigenvalues by the standard method leads to tedious computations. To compute them efficiently, one can proceed as follows. For each nonzero eigenvalue λ , consider a corresponding eigenvector $v = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$. Use the linear dependence between the rows of A to derive relations between a , b , c , and d . Then, write $Av = \lambda v$ explicitly and solve the resulting linear system.

$$\begin{pmatrix} 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 5 & 5 & 5 & 5 \\ 7 & 7 & 7 & 7 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 4 & 4 \\ 1 & 1 & 4 & 4 \\ 2 & 2 & 3 & 3 \\ 2 & 2 & 3 & 3 \end{pmatrix}.$$